

Mustafa Kerem Kurban

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Machine Learning Engineer | Data Scientist | MLOps & LLM Specialist | Neuroscientist

Innovative ML Engineer with 8+ years' experience designing and deploying large-scale ML & data pipelines, integrating big data technologies, and productionizing models for real-world impact. Skilled at collaborating in cross-functional teams to deliver batch and streaming pipelines, optimize partner workflows, and leverage GenAI for operational efficiency. Strong background in cloud-based systems, ML lifecycle management, and advanced data engineering for high-throughput environments. Holding a Swiss working permit (B) with full-time employment capacity.

Work Experience

Freelancer

Machine Learning Consultant

03/2025 - Current

- Conducted research on Large Language Models, focusing on **transformer optimization** techniques (GaLoRE, QLoRA, MoE) with an emphasis on limited GPU constraints.
- Possesses expertise in **Vision-Language Models** (VLMs) and **Vision Transformers** (ViTs), applied to image segmentation, video processing, and multimodal embeddings.
- Implemented **Reinforcement Learning** methodologies (SFT, RLHF, DPO) to enhance LLM performance using user-generated data.

EPFL Blue Brain Project, Geneva, Switzerland

Machine Learning Engineer

04/2024-12/2024

- Developed multi-agent LLM systems using LangChain and OpenAI Swarm for query-based scientific automation, **reducing manual analysis time by 90%**.
- Designed and deployed retrieval-augmented generation (RAG) pipelines, leveraging Neo4j for semantic search and personalized knowledge graph-based recommendations, **reducing hallucinations by >90%**.
- Designed semantic analysis pipelines to deliver **personalized insights for CRM and trend forecasting**, demonstrating cross-domain adaptability.
- Optimized AI workflows with AWS SQS and vector databases, ensuring real-time updates and scalability for production environments.

EPFL Blue Brain Project, Geneva, Switzerland

Scientific Software Developer | Data Scientist

04/2021-04/2024

- Architected high-throughput **simulation** and **analytics** pipelines (>100TB multimodal data) using Slurm-based HPC orchestration, Parquet, and HDF5 for distributed processing and storage.
- **Automated** large-scale graph analytics pipelines using Luigi to orchestrate DAGs of tasks on Slurm-based HPC systems, integrating PyTorch Geometric, **PySpark**, Pandas, and sparse array formats for feature extraction. Optimized low-level matrix multiplications to accelerate computations, enabling efficient multi-modal data processing at scale.

For open sourced projects, publications and additional details, visit keremkurban.created.app.

Education

BSc in Life Sciences and Engineering focusing on Neuroimaging and Deep Learning, Bogazici University, Turkiye

MSc in Engineering focusing on Computer Science and Neuroscience, Bilkent University, Turkiye & EPFL, Switzerland

Certificates

- Neo4j Certified Professional (2024)
- LLMOps and Fine Tuning Large Language Models (DeepLearning.AI, 2024/25)
- AI Agents in LangGraph (DeepLearning.AI, 2024)
- LLMs as Operating Systems: Agent Memory (DeepLearning.AI, 2025)
- Federated Fine tuning of LLMs with Private Data (DeepLearning.AI, 2025)
- AWS CDA DVA-C02 Certification (In progress), AWS Foundations, AWS Bedrock (AWS, 2024)
- Infrastructure as Code in Google Cloud Platform (Linkedin Learning, 2024)
- Quantization Fundamentals with Hugging Face (DeepLearning.AI, 2025)

Technical Skills

- **Programming Languages:** Python, Bash, MATLAB
- **AI and Data Tools:** PyTorch, TensorFlow, LangChain, Scikit-learn, Hugging Face Transformers, Unsloth, Peft, MCP
- **Caching & Databases:** Neo4j, SQL, NoSQL, Elasticsearch, Opensearch, Redis
- **Cloud Solutions:** AWS (S3, EC2, ELB, RDS, Route53, VPC, CloudFront, ECS, ECR, Fargate, SageMaker, Bedrock, Lambda, CloudWatch, CloudFormation, Beanstalk), Neo4j Aura DB/DS; experience with GCP and Azure.
- **DevOps Tools:** Docker, Kubernetes, Git, Terraform
- **Graph Analytics:** NetworkX, GraphSAGE, Cypher Query Language, GraphQL
- **Fullstack Development:** React.js, HTML/CSS, REST APIs, FastAPI
- **Languages:** English (C1), French (B1), Japanese (B1), Turkish (Native)

Projects

Knowledge Graph Powered Chat Agent with Neo4j and LangChain:

Designed a semantic analysis pipeline integrating Neo4j and RAG to process and personalize CRM workflows for **50k+ customer records**, reducing response time by **60%**. Leveraged AWS for scalability and ensured compliance with data privacy standards. (see <https://github.com/BlueBrain/citation-graph>)

Agentic Workflows for Simulation Automation:

Developed a human-in-the-loop AI system using LangChain and OpenAI APIs to automate NEURON simulation workflows. Enhanced efficiency by **70%**, achieving seamless integration with existing HPC infrastructure. (<https://github.com/BlueBrain/neuroagent>)

Front and Backend Development for Custom Chatbot Interfaces:

Developed a personalized website using React, Node.js, and CSS, featuring blog sections with keyword-based filtering to enhance UX, and **designed a custom Chat UI** to optimize usability for in-house chatbot applications tailored to Graph RAG application.

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